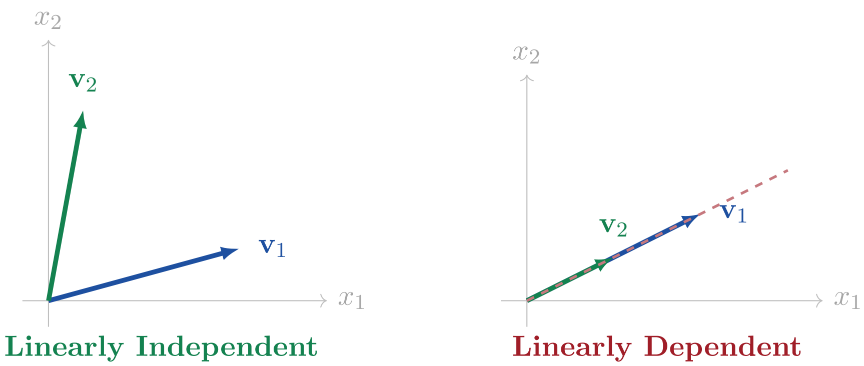
Linear Algebra
Linear Independence, Basis & Dimension

A Comprehensive Mathematical Treatment

Covering linear independence and dependence,
bases of vector spaces, the dimension theorem,
and the Rank–Nullity theorem.



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1 Introduction

The notions of **linear independence**, **basis**, and **dimension** are the cornerstones of linear algebra. Together they allow us to measure, in a precise algebraic sense, the "size" of a vector space and to identify the most economical sets of vectors that completely describe it.

Building on the theory of vector spaces and subspaces from the previous chapters, this chapter answers three fundamental questions:

- When is a set of vectors *redundant* in the sense that one of them can be expressed as a combination of the others?
- What is the smallest spanning set for a vector space?
- How many vectors are in every such minimal spanning set?

2 Linear Independence

2.1 Definition

Definition 2.1: Linear Independence and Dependence

Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in a vector space V. The vectors are said to be **linearly dependent** if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$, *not all zero*, such that

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = \mathbf{0}.$$

If the *only* solution to the above equation is $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$, the vectors are said to be **linearly independent**.

Remark 2.1: Intuitive Interpretation

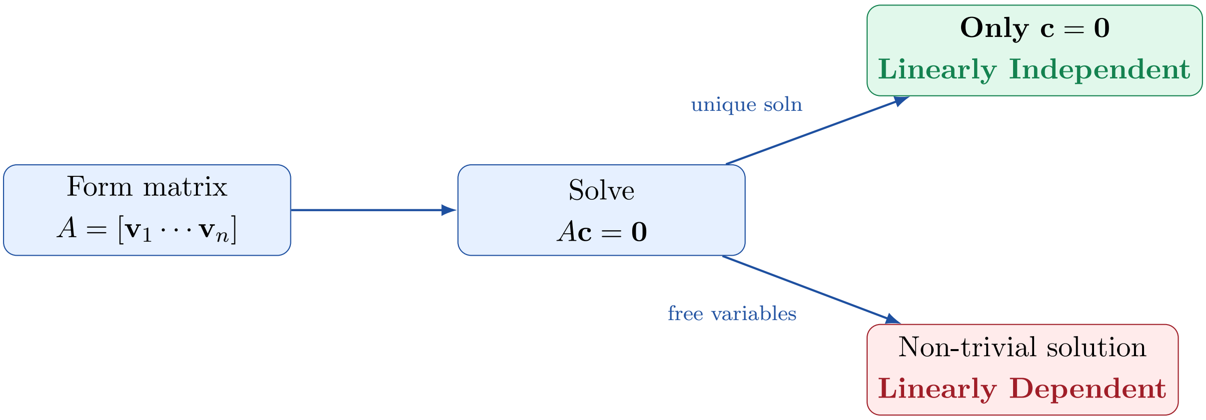
Linear dependence means that at least one vector in the set is *redundant*: it can be written as a linear combination of the others. Concretely, if $\alpha_1 \neq 0$ in a non-trivial relation, then

$$\mathbf{v}_1 = -\frac{\alpha_2}{\alpha_1}\mathbf{v}_2 - \cdots - \frac{\alpha_n}{\alpha_1}\mathbf{v}_n.$$

Linear independence, by contrast, means no vector in the set can be expressed this way; every vector contributes a genuinely new "direction".

2.2 Testing Linear Independence

To test whether $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbb{R}^m$ are linearly independent, form the homogeneous system with coefficient matrix $A = [\,\mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_n\,]$ and solve $A\mathbf{c} = \mathbf{0}$:



2.3 Examples in $\mathbb{R}^n$

Example 2.1: Linearly independent vectors in $\mathbb{R}^3$

Determine whether $\mathbf{v}_1 = (1, 0, 2)^T$, $\mathbf{v}_2 = (2, 1, 0)^T$, $\mathbf{v}_3 = (0, 1, -4)^T$ are linearly independent.
Form $A = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$ and row-reduce:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & -4 \end{pmatrix} \xrightarrow{R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -4 & -4 \end{pmatrix} \xrightarrow{R_3 + 4R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The system $A\mathbf{c} = \mathbf{0}$ has a free variable ($c_3 = t$), giving the non-trivial solution $\mathbf{c} = t(-2, -1, 1)^T$. Therefore the vectors are **linearly dependent**. Indeed:

$$-2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} -2 \\ 0 \\ -4 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ -8 \end{pmatrix} \neq \mathbf{0}.$$

(Scaling by $-\frac{1}{2}$: $\mathbf{v}_1 + \frac{1}{2}\mathbf{v}_2 - \frac{1}{2}\mathbf{v}_3 = \mathbf{0}$, confirming dependence.)

Example 2.2: Linearly independent vectors in $\mathbb{R}^3$

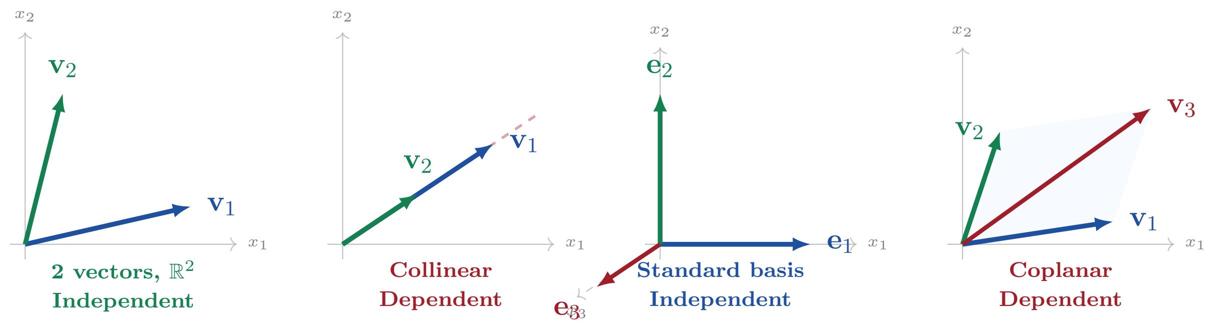
Determine whether $\mathbf{u}_1 = (1, 0, 0)^T$, $\mathbf{u}_2 = (0, 1, 0)^T$, $\mathbf{u}_3 = (0, 0, 1)^T$ are linearly independent.
The system $A\mathbf{c} = \mathbf{0}$ with $A = [\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{u}_3] = I_3$ reduces immediately:

$$I_3\mathbf{c} = \mathbf{0} \implies \mathbf{c} = \mathbf{0}.$$

The only solution is the trivial one, so the vectors are **linearly independent**.

2.4 Geometric Interpretation

In $\mathbb{R}^2$ and $\mathbb{R}^3$, linear (in)dependence has a clear geometric meaning:



2.5 Key Theorems on Linear Independence

Theorem 2.1: Size Bound for Independent Sets

If V is spanned by n vectors, then any set of more than n vectors in V must be linearly dependent.

Proof. Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ span V and let $\{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ be any set in V with $m > n$. Write each $\mathbf{w}_j = \sum_{i=1}^{n} a_{ij}\mathbf{u}_i$. Consider $\sum_{j=1}^{m} c_j \mathbf{w}_j = \mathbf{0}$; substituting gives $\sum_{i=1}^{n} \big(\sum_{j=1}^{m} a_{ij}c_j\big)\mathbf{u}_i = \mathbf{0}$. Since $m > n$, the homogeneous system $A\mathbf{c} = \mathbf{0}$ (with A an $n \times m$ matrix) has a non-trivial solution, yielding a non-trivial dependence relation. ∎

Theorem 2.2: Dependence and Span

A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent if and only if at least one vector in the set belongs to the span of the preceding ones:

$$\exists\, k \in \{2, \ldots, n\} : \mathbf{v}_k \in \operatorname{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}).$$

Proof. ($\Rightarrow$) Suppose $\sum_{i=1}^{n} \alpha_i \mathbf{v}_i = \mathbf{0}$ with some $\alpha_k \neq 0$. Choose the largest such k. Then $\mathbf{v}_k = -\alpha_k^{-1} \sum_{i<k} \alpha_i \mathbf{v}_i \in \operatorname{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$.

($\Leftarrow$) If $\mathbf{v}_k = \sum_{i=1}^{k-1} \beta_i \mathbf{v}_i$, then $\beta_1\mathbf{v}_1 + \cdots + \beta_{k-1}\mathbf{v}_{k-1} + (-1)\mathbf{v}_k + 0 \cdot \mathbf{v}_{k+1} + \cdots = \mathbf{0}$ is a non-trivial dependence relation. ∎

Example 2.3: Linear independence in P_3

Show that $\{1,\, x,\, x^2\}$ is linearly independent in P_3.

Suppose $\alpha_0 \cdot 1 + \alpha_1 \cdot x + \alpha_2 \cdot x^2 = 0$ for all $x \in \mathbb{R}$. Evaluating at $x = 0,\, 1,\, -1$ gives the system:

$$\alpha_0 = 0, \quad \alpha_0 + \alpha_1 + \alpha_2 = 0, \quad \alpha_0 - \alpha_1 + \alpha_2 = 0,$$

which yields $\alpha_0 = \alpha_1 = \alpha_2 = 0$. Hence $\{1, x, x^2\}$ is linearly independent.

3 Basis of a Vector Space

3.1 Definition

Definition 3.1: Basis

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in a vector space V is called a **basis** for V if it satisfies both:

(i) $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ **spans** V; that is, $\text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n) = V$.

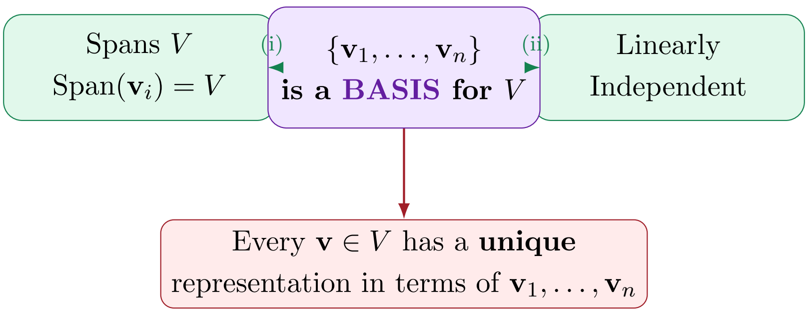
(ii) $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is **linearly independent**.

Equivalently, $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis if and only if every vector $\mathbf{v} \in V$ can be written *uniquely* as a linear combination $\mathbf{v} = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$.

Remark 3.1: Uniqueness of Coordinates

The scalars $\alpha_1, \ldots, \alpha_n$ in the unique representation $\mathbf{v} = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$ are called the **coordinates of v with respect to the basis** $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, written $[\mathbf{v}]_{\mathcal{B}} = (\alpha_1, \ldots, \alpha_n)^T$.

3.2 The Equivalence: Spanning + Independence = Basis



3.3 Standard Bases

Example 3.1: Standard basis for $\mathbb{R}^n$

The set $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ is the vector with a 1 in position i and 0s elsewhere, is the **standard basis** for $\mathbb{R}^n$:

$$\mathbf{e}_1 = \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix}0\\1\\\vdots\\0\end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix}0\\0\\\vdots\\1\end{pmatrix}.$$

For any $\mathbf{x} = (x_1, \dots, x_n)^T \in \mathbb{R}^n$: $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$ (spanning), and the only solution to $\sum \alpha_i \mathbf{e}_i = \mathbf{0}$ is $\alpha_i = 0$ for all i (independence).

Example 3.2: Standard basis for P_n

The set $\{1, x, x^2, \dots, x^{n-1}\}$ is the standard basis for P_n. Every $p \in P_n$ writes uniquely as $p(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}$, giving coordinates $(a_0, a_1, \dots, a_{n-1})^T$.

Example 3.3: Standard basis for $\mathbb{R}^{m\times n}$

For the space $\mathbb{R}^{m\times n}$, define E_{ij} to be the $m \times n$ matrix with a 1 in position (i, j) and 0s elsewhere. The set $\{E_{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is a basis for $\mathbb{R}^{m\times n}$. For example, the standard basis of $\mathbb{R}^{2\times 2}$ is:

$$E_{11} = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}, \quad E_{12} = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}, \quad E_{21} = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix}, \quad E_{22} = \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}.$$

3.4 Existence and Reduction of Spanning Sets

Theorem 3.1: Spanning Set Contains a Basis

Every finite spanning set for a non-zero vector space V contains a basis.

Proof. Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ span V. If S is linearly independent, it is already a basis. Otherwise, by Theorem 2.2, some $\mathbf{v}_k \in \mathrm{Span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$, so removing $\mathbf{v}_k$ leaves a set that still spans V. Repeat until the remaining set is linearly independent; the result is a basis. ∎

Theorem 3.2: Extending Independent Sets to a Basis

Every linearly independent subset of a finite-dimensional vector space V can be extended to a basis for V.

3.5 Worked Example: Finding a Basis from a Spanning Set

Example 3.4: Basis for a subspace of $\mathbb{R}^4$

Find a basis for $W = \operatorname{Span}\big((1,2,1,0)^T,\ (2,4,2,0)^T,\ (1,0,1,2)^T,\ (0,2,0,-2)^T\big)$. Place the vectors as rows and row-reduce:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 1 & 0 & 1 & 2 \\ 0 & 2 & 0 & -2 \end{pmatrix} \xrightarrow[R_3-R_1]{R_2-2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 2 \\ 0 & 2 & 0 & -2 \end{pmatrix} \xrightarrow{R_4+R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The two non-zero rows correspond to the first and third original vectors (rows 1 and 3 of the original matrix are the pivot rows). A basis for W is:

$$\big\{(1,2,1,0)^T,\ (1,0,1,2)^T\big\}.$$

4 Dimension of a Vector Space

4.1 The Dimension Theorem

Theorem 4.1: All Bases Have the Same Size

Let V be a vector space. If $\{\mathbf{u}_1, \ldots, \mathbf{m}\}$ and $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ are both bases for V, then $m = n$.

Proof. Since $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ spans V and $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is linearly independent, Theorem 2.1 gives $n \leq m$. Symmetrically, $m \leq n$. Hence $m = n$. ■

This theorem justifies the following definition.

Definition 4.1: Dimension

Let V be a vector space. If V has a basis consisting of exactly n vectors, we say V is **finite-dimensional** and define the **dimension** of V to be

$$\dim(V) = n.$$

The zero vector space $\{\mathbf{0}\}$ is defined to have dimension 0. If no finite basis exists, V is called **infinite-dimensional**.

4.2 Dimensions of Common Spaces

Space	Standard Basis	Dimension	Notes
$\mathbb{R}^n$	$\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$	n	
$\mathbb{R}^{m\times n}$	$\{E_{ij}\}$, $1 \le i \le m$, $1 \le j \le n$	mn	
P_n	$\{1, x, x^2, \dots, x^{n-1}\}$	n	
$C[a, b]$	No finite basis	∞	Infinite-dimensional
$\{\mathbf{0}\}$	Empty set	0	Zero subspace

4.3 Dimension and Subspaces

Theorem 4.2: Dimension of Subspaces

Let S be a subspace of a finite-dimensional vector space V. Then:

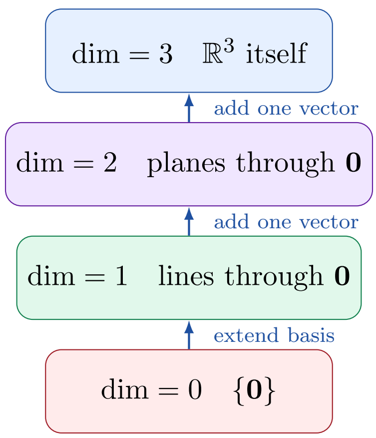
(i) $\dim(S) \le \dim(V)$.

(ii) $\dim(S) = \dim(V)$ if and only if $S = V$.

Proof. **(i)** Any basis of S is a linearly independent set in V. By Theorem 2.1, its size cannot exceed $\dim(V)$.

(ii) If $\dim(S) = \dim(V) = n$, take any basis $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ of S. Being linearly independent and of size $n = \dim(V)$, it is also a basis for V (by Theorem 3.2), so $S = \text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_n) = V$. ■

The diagram below shows how dimension stratifies the subspaces of $\mathbb{R}^3$:



4.4 A Practical Characterisation

Theorem 4.3: Three Equivalent Conditions

Let V be an n-dimensional vector space and let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be any set of exactly n vectors in V. Then the following are equivalent:

(i) S is a basis for V.

(ii) S spans V.

(iii) S is linearly independent.

Proof. **(i)⇒(ii) and (i)⇒(iii)** follow directly from the definition of basis.

(ii)⇒(i) If S spans V, it contains a basis (Theorem 3.1). That basis has $n = \dim(V)$ vectors, and since $|S| = n$, removing any vector would make $|S| < n$, so S itself is the basis.

(iii)⇒(i) If S is linearly independent, extend it to a basis B (Theorem 3.2); since $|B| = n = |S|$, we have $B = S$, so S is a basis. ■

5 Rank, Nullity, and the Rank–Nullity Theorem

5.1 Row Space, Column Space, and Rank

Definition 5.1: Row Space and Column Space

Let A be an $m \times n$ matrix.

- The **row space** of A, denoted $\mathrm{Row}(A)$, is the span of the row vectors of A, viewed as a subspace of $\mathbb{R}^n$.
- The **column space** of A, denoted $\mathrm{Col}(A)$, is the span of the column vectors of A, viewed as a subspace of $\mathbb{R}^m$.
- The **rank** of A is $\mathrm{rank}(A) = \dim(\mathrm{Row}(A)) = \dim(\mathrm{Col}(A))$.
- The **nullity** of A is $\mathrm{nullity}(A) = \dim(N(A))$.

5.2 The Rank–Nullity Theorem

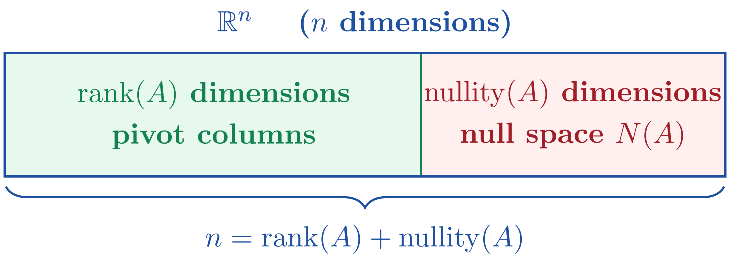
Theorem 5.1: Rank–Nullity Theorem

For any $m \times n$ matrix A,

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = n.$$

Proof. Let $r = \operatorname{rank}(A)$. Row-reduction of A produces r pivot columns and $n - r$ free columns. The $n - r$ free variables parametrise $N(A)$; the corresponding $n - r$ special solutions form a basis for $N(A)$, so $\operatorname{nullity}(A) = n - r$. Therefore $r + (n - r) = n$. ■

The Rank–Nullity Theorem can be visualised as a partition of $\mathbb{R}^n$:



5.3 Worked Example

Example 5.1: Rank, nullity and the theorem

Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Row-reduce:

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are 2 pivot columns (columns 1 and 3), so $\operatorname{rank}(A) = 2$. The free variables are x_2 and x_4, so $\operatorname{nullity}(A) = 2$. Indeed: $\operatorname{rank}(A) + \operatorname{nullity}(A) = 2 + 2 = 4 = n$. ✓
A basis for $N(A)$ (setting $(x_2, x_4) = (1, 0)$ then $(0, 1)$):

$$N(A) = \operatorname{Span}\left(\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right).$$

6 Summary and Connections

Chapter Summary

- Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are **linearly independent** iff $\sum \alpha_i \mathbf{v}_i = \mathbf{0} \Rightarrow$ all $\alpha_i = 0$. Otherwise they are **linearly dependent** and some vector is redundant.
- A **basis** for V is a linearly independent spanning set. Every vector in V has a *unique* coordinate representation in any basis.
- All bases of a finite-dimensional vector space V have the same number of elements; this common count is $\dim(V)$.
- Any $\dim(V)$ vectors that span V, or that are linearly independent, automatically form a basis.
- For an $m \times n$ matrix A: $\text{rank}(A) + \text{nullity}(A) = n$ (Rank–Nullity Theorem).

The diagram below summarises the relationships among all key concepts.